YIHAO (BEN) ZHOU

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EDUCATION

University of California, Berkeley — Berkeley, CA
Intended Majors: Data Science, Physics, Applied Math — *Aug 2025 – May 2029*

- Linear Algebra & Differential Equations, Computational Structures in Data Science, Intro to General Astronomy, Philo of Math

Phillips Academy Andover — Andover, MA
5.83 GPA (4.17/4.0 UW, 1st decile) 1570 SAT — *Sep 2021 – Jun 2025*

- Calculus BC, Physics C: Mechanics E&M, Computer Science, Multivariable Calculus, Linear Algebra, Fluid Mechanics
- Awards: John Locke Institute 2024 Global Essay Commendation award in Philosophy (Global top 16.9%), Harvard International Review Outstanding Content Award, World Astrophotography Youth Competition 1st Place; Starryscape Optolong 2024 Shortlist.

RESEARCH EXPERIENCE

Composition and Properties of Comet C/2020 F3 (NEOWISE)'s tail — Flagstaff, AZ
Research on comet NEOWISE with University of Maryland Professor Quanzhi Ye — *Sep 2022 – Jun 2024*

- Created and analyzed computerized mosaic image of comet, determined 3 tails made from ions, sodium, & dust
- Presented results at the Asteroids, Comets, and Meteors Conference 2023 in Flagstaff, Arizona
- Published in Research Notes of the American Astronomical Society, created a Computer model of NEOWISE's tail using sbpy/Finson-Probstein model, stacked and processed images using SWarp and Python

Pioneer Research Program — Remote
Relativity, Cosmology, and the Universe with JHU Professor David Kaplan — *Feb 2024 – Jul 2024*

- Authored 12-page paper: Investigating Attainable Velocities w/ Current Material and Laser Capabilities for Breakthrough Starshot

EXTRACURRICULAR ACTIVITIES

Boomerang | COO, and CTO of Startup building an AI Co-founder and evaluation pipeline | International — *Apr 2025 – Present*

- Made multi-agent RAG + quantitative scoring and automation engine orchestrated with LangChain, n8n, and HTML/CSS/JS.
- Raised 60,000 grant funding from Thai Summit Group; Endorsed by BNP Paribas, LINE MAN Wongnai; Helped advertise and create content that logged 112,000+ impressions and 12k+ followers across platforms. Created pitch decks.
- Built scoring core w/ PyTorch regression head + HF (0–100). Including per-segment CAGR vector, capital scoring, competitor research, sentiment evaluation, ETF–SPY spread. Agentic powered Bloomberg Terminal eval pipelines to validate startup ideas.

Quantitative trader | Online — *Jan 2025 – Present*

- Used Python, C++, and QuantConnect Quantbook to develop and live test trading strategies.
- Employed Monte Carlo permutation tests and Conditional Parameter Optimizations to fine-tune strategies.
- Processed model dataset using outlier identification, min-max scaling, z-score normalization, ADF tests/fractional differentiation for stationarity, training/test/walk-forward set preparation.
- Built Moving Average Reversion strategy on the currency pair USDCHF with 25% CAR and 1.12 Sharpe ratio Prev. 400k Funded at FTMO; US Equities strategy with a Regression model predicting the effect of Positive-Negative splits with a 0.8 Sharpe.

Starry Sky Check-in Challenge | Ambassador and Coordinator | International — *May 2024 – Aug 2025*

- Provided astronomy education to residents in underprivileged areas to use astronomy as a tourism resource to drive economic growth. Project endorsed by the Office of Astronomy Development of the International Astronomical Union
- Created curriculum, made videos, and taught, generating 16,000 USD in revenue, 1M+ views/website visits

Astrolink | Founder of global non-profit offering remote telescope access | Andover, MA — *Mar 2024 – Present*

- Partnered with 35+ schools and astronomy clubs in 6 states (MA, NY, OR, NJ, CT, and CA)
- Collaborated with the National Science Teaching Association, reached 1000+ students in classrooms, and helped create 7 student-run affiliate clubs

T CrB light curve modeling | Intern for Young Astronomy Teachers Link | Remote — *Apr 2024 – Sep 2024*

- Worked with Tsinghua University students to conduct light measurements and created a tutorial for observation of the pre-eruption dip of T CrB for the national Young Astronomy Teachers Link
- Analyzed and created a data curve using Python Pandas, submitted data to the American Association of Variable Star Observers

Varsity Swimming | Varsity swimmer at 3rd-ranked swimming high school | International — *Sep 2011 – Present*

- Recognized as NISCA All-American in 2024 (US top 100 in an event); Division 1 All-NEPSAC in 2025 (New England Top 8 in an Event); Easterns/New England champion 2023, 2024, 2025

ADDITIONAL INFORMATION

Languages: Chinese (Native), English (Native), German (Intermediate)
Computer Skills: Python, SQL, C++, TS, Html/CSS/JS, Final cut pro/Premiere pro, Java, Excel
Hobbies: Astrophotography, Basketball, GeoGuessr, Oil painting, Reading, Poker